Calculus

Techniques for find antiderivatives of more complicated functions as fuck

8/20/2026 Lecture

Integration of multiplied functions

Due to integration is antidifferentiation (i.e. opposite direction of differentiation), integrating each multiplied function separately and then multiplying them couldn't work at all.

$$\int x \ln x \, dx \neq \int x dx \cdot \int \ln x \, dx$$

Then, how do we integrate that?
Assume that the Integrand is product of two functions, and one of them is derivatives of some function. Then, by the product rules,

$$\frac{d}{dx} f(x) \cdot g(x) = \frac{df(x)}{dx} \cdot g(x) + f(x) \cdot \frac{dg(x)}{dx}$$

$$\rightarrow f(x) \cdot \frac{dg(x)}{dx} = \frac{d}{dx} f(x) \cdot g(x) - \frac{df(x)}{dx} \cdot g(x)$$

$$\therefore \int \left\{ f(x) \cdot \frac{dg(x)}{dx} \right\} dx = f(x) \cdot g(x) - \int \left\{ \frac{df(x)}{dx} \cdot g(x) \right\} dx$$

Or simply,

$$\int f(x) \cdot g'(x) dx = f(x) \cdot g(x) - \int f'(x) \cdot g(x) dx$$

Via using above relation, integration of multiple functions is simply calculated.

$$\int x \ln x \, dx = \int \left(\frac{x^2}{2} \right)' \ln x \, dx = \frac{x^2}{2} \ln x - \frac{1}{2} \int x^2 \cdot (\ln x)' \, dx$$

During derivative of ln(x) = 1/x, second term of right side simply calculated as integration of x.

Techniques – Integration by parts

For multiplied function,

$$f(x) \cdot g'(x)$$

Its anti-derivative will be

$$\int f(x) \cdot g'(x)dx = f(x) \cdot g(x) - \int f'(x) \cdot g(x)dx$$

Exercises 1:

Express the process of finding anti-derivatives of given function:

$$h(x) = 3f(x) \cdot g'(x)$$

Exercises 2:

Find anti-derivatives of given function:

$$f(x) = xe^x$$

Exercises 3:

Find anti-derivatives of given function:

$$f(x) = \ln x$$

Hint) Sometimes, simple digits can be a derivative of function!

Exercises 4:

Find anti-derivatives of given function:

$$f(x) = e^x \sin x$$

Hint) There is me inside me!!

Integration of fraction

Fractional form is also hard to integrate. Some kind of functions can be solved by substitution, but most of functions are like below,

$$\int \frac{1}{x^2 + 2x - 3} dx$$

If function have nth degree polynomial in its denominator, we can express that to sum of two or more fractions.

For instance,

$$\frac{1}{x^2 + 2x - 3} = \frac{A}{(x+3)} + \frac{B}{(x-1)}$$

$$B(x+3) + A(x-1) = x^2 + 2x - 3 \rightarrow A = \frac{1}{4}, B = -\frac{1}{4}$$

$$\therefore \frac{1}{x^2 + 2x - 3} = \frac{1}{4(x+3)} - \frac{1}{4(x-1)}$$

$$\rightarrow \int \left\{ \frac{1}{4(x+3)} - \frac{1}{4(x-1)} \right\} dx = \frac{1}{4} \ln \left| \frac{x+3}{x-3} \right| + C$$

Techniques – partial fraction

$$\frac{c}{AB} = \frac{c}{B-A}\left(\frac{1}{A} - \frac{1}{B}\right) (while\ c = constant)$$

$$\frac{N(x)}{(ax+b)(cx+d)} \equiv \frac{A}{(ax+b)} + \frac{B}{(cx+d)}$$

$$\frac{N(x)}{(ax+b)^2} \equiv \frac{A}{(ax+b)} + \frac{B}{(ax+b)^2}$$

$$\frac{N(x)}{(ax+b)(x^2+c)} \equiv \frac{A}{(ax+b)} + \frac{Bx+c}{(x^2+c)}$$

(sorry)

Exercises 1:

Find anti-derivatives of given function:

$$f(x) = \frac{1}{x^2 - 2x + 4}$$

Exercises 2:

Find anti-derivatives of given function:

$$f(x) = \frac{1}{(x - 7)(x^2 + 9)}$$

Integration of complex Trigonometric functions

You can imagen that the differentiation of trigonometric functions.
Consider the integration is calculation have opposite directions to differentiation, we can easily find the result.

For example, from differentiation of sin(x),

$$\frac{d}{dx}\sin x = \cos x$$

$$d\sin x = \cos x \; dx$$

$$\sin x + c = \int \cos x \; dx$$

We can recognize that the integration of cos(x) is sin (x).
Integration of sin(x) is also found easily via using differentiation of cos(x).
How about product of trigonometric function?

$$\int \cos 3x \sin 4x \, dx$$

To Integrate multiplied functions, it must be separated as the sum of two or more functions.
Via using relationship below,

$$\cos A \sin B = \frac{1}{2}\{\sin(A+B) - \sin(A-B)\}$$

$$\int \cos 3x \sin 4x \, dx = \frac{1}{2}\int \{\sin(7x) - \sin(-x)\} \, dx$$

It simply expressed to sum of two functions.

Techniques – Angle addition and subtraction formulas

For trigonometric function,

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$$

$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$$

$$\sin\alpha\cos\beta = \frac{1}{2}[\sin(\alpha+\beta) + \sin(\alpha-\beta)]$$

$$\cos\alpha\cos\beta = \frac{1}{2}[\sin(\alpha+\beta) - \sin(\alpha-\beta)]$$

$$\cos\alpha\cos\beta = \frac{1}{2}[\cos(\alpha+\beta) + \cos(\alpha-\beta)]$$

$$\sin\alpha\sin\beta = -\frac{1}{2}[\cos(\alpha+\beta) - \cos(\alpha-\beta)]$$

Exercises 1:

Find anti-derivatives of given function:

$$f(x) = \cos 5x \sin 8x$$

Exercises 2:

Find anti-derivatives of given function:

$$f(x) = \sin 5x \sin 8x$$

Exercises 3:

Find anti-derivatives of given function:

$$f(x) = (\cos 3x)^2$$

***Useful but not on the lecture**

Trigonometric – reduction formular

Techniques – integration of $\sin^n x$ on $x \geq 2$

$$\int (\sin x)^n \, dx = -\frac{1}{n}(\sin x)^{n-1} \cos x + \frac{(n-1)}{n} \int (\sin x)^{n-2} \, dx \; (n \geq 2)$$

Pf.

$$let \; I_n = \int (\sin x)^n \, dx$$

$$\int (\sin x)^n \, dx = \int \sin x \cdot (\sin x)^{n-1} \, dx$$

$$via \; using \int f(x) \cdot g'(x) \, dx = f(x)g(x) + \int f'(x) \cdot g(x) \, dx$$

$$\rightarrow \int \sin x \cdot (\sin x)^{n-1} \, dx = -(\sin x)^{n-1} \cos x + (n-1) \int (\sin x)^{n-2} (\cos x)^2 \, dx$$

$$the \; second \; term: (n-1) \int (\sin x)^{n-2} (\cos x)^2 \, dx$$

$$= (n-1) \left\{ \int (\sin x)^{n-2} \, dx - \int (\sin x)^n \, dx \right\}$$

$$= (n-1) \left\{ \int (\sin x)^{n-2} \, dx - I_n \right\}$$

$$\therefore \; nI_n = -(\sin x)^{n-1} \cos x + (n-1) \int (\sin x)^{n-2} \, dx,$$

$$I_n = -\frac{1}{n}(\sin x)^{n-1} \cos x + \frac{(n-1)}{n} \int (\sin x)^{n-2} \, dx \; (n \geq 2)$$

Exercises 1:

Find anti-derivatives of given function:

$$f(x) = (\sin x)^3$$

Techniques – integration of $\cos^n x$ on $x \geq 2$

$$\int (\cos x)^n \, dx = \frac{1}{n}(\cos x)^{n-1} \sin x + \frac{(n-1)}{n} \int (\cos x)^{n-2} \, dx \; (n \geq 2)$$

Pf.

$$let \; I_n = \int (\cos x)^n \, dx$$

$$\int (\cos x)^n \, dx = \int \cos x \cdot (\cos x)^{n-1} \, dx$$

$$via \; using \int f(x) \cdot g'(x) \, dx = f(x)g(x) + \int f'(x) \cdot g(x) \, dx$$

$$\rightarrow \int \cos x \cdot (\cos x)^{n-1} \, dx = (\cos x)^{n-1} \sin x + (n-1) \int (\cos x)^{n-2} (\sin x)^2 \, dx$$

$$the \; second \; term: (n-1) \int (\cos x)^{n-2} (\sin x)^2 \, dx$$

$$= (n-1) \left\{ \int (\cos x)^{n-2} \, dx - \int (\cos x)^n \, dx \right\}$$

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$$\therefore \; nI_n = (\cos x)^{n-1} \sin x + (n-1) \int (\cos x)^{n-2} \, dx,$$

$$I_n = \frac{1}{n}(\cos x)^{n-1} \sin x + \frac{(n-1)}{n} \int (\cos x)^{n-2} \, dx \; (n \geq 2)$$

Exercises 1:

Find anti-derivatives of given function:

$$f(x) = (\cos x)^3$$